PULLBACKS OF MEROMORPHIC FUNCTIONS ARE DEFINED IN SEVEN SITUATIONS RANGING FROM FLATNESS TO WEAK ASSOCIATION ASSUMPTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-pullback-meromorphic-sections-defined is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 5983-6004 and proof span 6006-6031 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement is a seven-item list; all seven cases are part of the counted statement span and the proof labels them by number exactly as the author does.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The characterization of regular sections of the structure sheaf, which reduces the existence of pullbacks of meromorphic functions to the requirement that the ring map send nonzerodivisors to nonzerodivisors. Cited at proof lines 6008-6011. Turns the sheaf-theoretic statement into the algebraic statement about nonzerodivisors.
- An element is a nonzerodivisor exactly when it lies outside every weakly associated prime. Cited at proof lines 6017-6020. Gives the main case, where the hypothesis on weakly associated points prevents the image of a nonzerodivisor from being a zerodivisor.
- The comparison between associated and weakly associated primes, and the description of weakly associated points of a reduced scheme, both cited from the same chapter. Cited at proof lines 6021-6026. Derives the locally Noetherian cases and the reduced case from the main case.
- Elements of a minimal prime of a ring are zerodivisors. Used at proof lines 6014-6016. Shows that a nonzerodivisor avoids every minimal prime, which is what the hypotheses on the images of the relevant points supply.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05C3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6906–6906. Same-chapter reference used in the retained author text at source line(s) 2624, 6025, 6106, 6127. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 05AR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6859–6859. Same-chapter reference used in the retained author text at source line(s) 875, 965, 1222, 2625, 6026, 6345. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01WZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2500–2500. Same-chapter reference used in the retained author text at source line(s) 2934, 6011. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0EME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18304–18304. Same-chapter reference used in the retained author text at source line(s) 2628, 6029, 6463. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. In each of the following cases pullbacks of meromorphic functions are defined.*

1. *every weakly associated point of X maps to a generic point of an irreducible component of Y,*
2. *X, Y are integral and f is dominant,*
3. *X is integral and the generic point of X maps to a generic point of an irreducible component of Y,*
4. *X is reduced and every generic point of every irreducible component of X maps to the generic point of an irreducible component of Y,*
5. *X is locally Noetherian, and any associated point of X maps to a generic point of an irreducible component of Y,*
6. *X is locally Noetherian, has no embedded points and any generic point of an irreducible component of X maps to the generic point of an irreducible component of Y, and*
7. *f is flat.*

Proof. The question is local on X and Y. Hence we reduce to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(R)$ and f is given by a ring map $\varphi : R \to A$. By the characterization of regular sections of the structure sheaf in Lemma 01WZ we have to show that $R \to A$ maps nonzerodivisors to nonzerodivisors. Let $t \in R$ be a nonzerodivisor.

If $R \to A$ is flat, then $t : R \to R$ being injective shows that $t : A \to A$ is injective. This proves (7).

In the other cases we note that t is not contained in any of the minimal primes of R (because every element of a minimal prime in a ring is a zerodivisor). Hence in case (1) we see that $\varphi(t)$ is not contained in any weakly associated prime of A.

Thus this case follows from Algebra, Lemma 05C3. Case (5) is a special case of (1) by Lemma 05AR. Case (6) follows from (5) and the definitions. Case (4) is a special case of (1) by Lemma 0EME. Cases (2) and (3) are special cases of (4). □